STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 7: Gravitation

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Kepler’s law of equal areas? [1]
2. Universal gravitation law? [1]
3. SI unit of G? [1]
4. g at surface of a spherical planet? [1]
5. Gravitational potential energy zero at infinity? [1]
6. Does escape speed depend on launched mass? [1]
7. What supplies centripetal force to a satellite? [1]
8. Kepler’s third law for fixed central mass? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. Force between 2 kg and 3 kg 1 m apart; G=6.67 x 10^-11. [2]
10. Weight of 50 kg for g=9.8? [2]
11. If orbit radius doubles, circular-speed factor? [2]
12. At altitude h=R, ratio g_h/g? [2]
13. Approximate potential-energy change for small height h? [2]
14. If orbital radius becomes 4r, period factor? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Find g at height h=R/2 relative to surface g. [3]
16. Escape speed for g=5 m/s^2 and R=2 x 10^6 m? [3]
17. GM=4 x 10^14, r=7 x 10^6 m: orbital speed? [3]
18. Weight is 600 N at surface. Weight at h=R? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Derive circular orbital speed and period. [4]
20. 1000 kg satellite at r=7 x 10^6 m, GM=4 x 10^14. Find speed and energy. [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Newton gravitation: F=Gm1m2/r^2; G=6.67 x 10^-11 N m^2 kg^-2.
- Surface gravity: g=GM/R^2; at radius r from the centre, g(r)=GM/r^2.
- At altitude h: g_h/g=(R/(R+h))^2; weight W=mg.
- Gravitational potential energy: U=-GMm/r; small-height change near surface Delta U approx mgh.
- Circular orbit: v=sqrt(GM/r); T=2pi sqrt(r^3/(GM)); hence T^2 proportional to r^3.
- Escape speed: v_e=sqrt(2GM/R)=sqrt(2gR).
- Circular orbit total energy: E=-GMm/(2r); orbital kinetic energy K=GMm/(2r).
- Kepler’s second law: equal areas are swept in equal times.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: The radius vector sweeps equal areas in equal times.
Final answer: The radius vector sweeps equal areas in equal times.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: F=Gm_1m_2/r^2.
Final answer: F=Gm_1m_2/r^2.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: N m^2 kg^-2.
Final answer: N m^2 kg^-2.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: g=GM/R^2.
Final answer: g=GM/R^2.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: U=-GMm/r.
Final answer: U=-GMm/r.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: No.
Final answer: No.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Gravity.
Final answer: Gravity.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: T^2 proportional to r^3.
Final answer: T^2 proportional to r^3.

Question 9

Working: F=Gm1m2/r^2=(6.67x10^-11)(2)(3)/(1^2)=4.002x10^-10 N, or 4.00x10^-10 N.
Final answer: 4.00 x 10^-10 N.

Question 10

Working: Weight W=mg=50(9.8)=490 N.
Final answer: 490 N.

Question 11

Working: v=sqrt(GM/r). If r becomes 2r, v_new/v_old=sqrt(r/(2r))=1/sqrt2.
Final answer: 1/sqrt2.

Question 12

Working: Altitude h=R makes the distance from the centre 2R. Since g is proportional to 1/r^2, g_h/g=R^2/(2R)^2=1/4.
Final answer: 1/4.

Question 13

Working: For a small height above the surface, Delta U=mgh, assuming g is approximately constant.
Final answer: DeltaU=mgh.

Question 14

Working: Kepler’s third law gives T proportional to r^(3/2). If radius becomes 4r, T_new/T_old=4^(3/2)=8.
Final answer: 8.

Question 15

Working: At h=R/2, the distance from centre is R+h=1.5R. Therefore g_h/g=(R/1.5R)^2=1/2.25=4/9.

Final answer: g_h/g=(2/3)^2=4/9.

Question 16

Working: v_escape=sqrt(2gR)=sqrt[2(5)(2x10^6)]=sqrt(2x10^7)=4.47x10^3 m/s.

Final answer: sqrt(2gR)=4.47 x 10^3 m/s.

Question 17

Working: v=sqrt(GM/r)=sqrt[(4x10^14)/(7x10^6)]=sqrt(5.714x10^7)=7.56x10^3 m/s.

Final answer: sqrt(GM/r)approx 7.56 x 10^3 m/s.

Question 18

Working: At h=R, gravity is one quarter of surface gravity. Weight is therefore 600/4=150 N.

Final answer: 150 N.

Question 19

Working: Set gravity equal to centripetal force: GMm/r^2=mv^2/r. Cancel m and solve: v=sqrt(GM/r). Then T=2pi r/v=2pi sqrt(r^3/GM).

Final answer: GMm/r^2=mv^2/r v=sqrt(GM/r); T=2pir/v=2pisqrt(r^3/GM).

Question 20

Working: v=sqrt(GM/r)=sqrt[(4x10^14)/(7x10^6)]=7.56x10^3 m/s. Total orbital energy E=-GMm/(2r)=-(4x10^14)(1000)/(14x10^6)=-2.86x10^10 J.

Final answer: vapprox 7.56 x 10^3 m/s; E=-GMm/(2r)approx -2.86 x 10^10 J.